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A concise approach to eigenform and reflexivity

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Abstract

Purpose – This paper aims to introduce the concept and praxis for eigenform in the context of second-order cybernetics.

Design/methodology/approach – The paper is designed as a formal (partly mathematical) introduction with excursions into the applications and meanings of these constructions. Mathematics studies what a distinction would be if there could be a distinction. Mathematics is a special form of fictional design. This study raises the question of "What it would mean to go beyond mathematics or for mathematics to go beyond itself?".

Findings – This study shows how objects in the author's experience can be seen to be eigenforms and that in this context such objects are a construct of their interactions, linguistic and otherwise experiential. In this way, the author can investigate scientifically without the need for an assumption of objectivity. The author cocreates the universe through the discovery of distinctions and eigenforms in their dialogue with what can be.

Research limitations/implications – The implications of this research are profound for the performance and exploration of science. The author can explore their role in that creation and find that what they create is independent of significant subsets of their actions.

Practical implications – The practical implications of this study are strongest for the logical understanding of the author's constructions and actions. They have used eigenform and reflexivity to maintain a clear view of their participation in their own worlds.

Social implications – The social implications are in accordance with the practical implications. The author can now admit that they each produce eigenform models of the others and for themselves. These models have in-depth usage, in that it is understood that one is not identical with their models.

Originality/value – This paper presents a highly original and very simple way to incorporate second-order cybernetics into all thought and action.

Keywords Cybernetics, Reflexivity, Circularity, Distinction, Eigenform, Reflexive domain

Paper type Research paper

Introduction to eigenform

An eigenform is a fixed point for a transformation. In this context, an arbitrary transformation is allowed. Transformation means change. To look for an eigenform is to look for something that does not change in the presence of change. We work with this notion all the time, making objects of ourselves, always changing and yet considered to be a single person, making constant objects of the moving display of our perceptions.

This usage of the notion of a fixed point is an extension of some technical uses of the term, but we mean it to be taken both informally and formally. In the next paragraph, I give a specific example of how an eigenform can arise as the fixed point of a transformation that is simple and syntactical. For our purposes, an eigenform is the analog of an eigenvector in an analysis or in linear algebra, but it is much more general and includes fixed points that occur in reflexive domains as will be explained in the following sections.

An important issue in this subject of eigenforms is the use of mathematics. I proceed in this paper to use a small amount of mathematics that is all based on the idea of making or representing a distinction. Mathematics is what comes from imagining that there could be a

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distinction and following the consequences of that imagination. In this view, there is no boundary between what anyone does with his or her mind and the language and mathematics. The marking of parentheses such as (these words in parentheses) is a mathematical form. What is required to follow and think about an eigenform is this exact level of mathematics and form. This does place a demand on the reader that he does not usually bring to consciousness. That demand is to ask you to actually look at the letters and notations in a text, or at the details in a perception, to really look, and I may make up rules about the distinctions in a text. This level of looking at written language is not a big demand to make, and it has deep conceptual rewards. For example, we can consider the following patterns of parentheses:

$$[], [[]], [[[]]], [[[[]]]], [[[[[]]]]], [[[[[[]]]]]], [[[[[[[]]]]]]], \ldots$$

Usually, when someone starts making patterns, we call it *design* or *artwork*. Here it is all the same. *Designs are made by working with distinctions and recursively (a word that all cyberneticians know) operating on previous designs to produce new designs*. What is mathematical here, but could just as well be called *patternplay*, is the observation of patterns in the forms produced and finding ways to articulate these forms. In this way, it is natural to define $T(x) = [x]$. Then I can apply T again and again to an arbitrary x as shown below:

$$\begin{gathered} x \\ [x] \\ [[x]] \\ [[[x]]] \\ [[[[x]]]] \\ [[[[[x]]]]] \\ \ldots \end{gathered}$$

If you do this, for a long time, it begins to look like:

$$E = [[[[[[[[[\ldots]]]]]]]]]$$

And this expression has the *form of a something that does not change* if you put one more set of brackets around it. Thus, E (above) is an eigenform for the transformation T where:

$$T(x) = [x]$$

The entity E appears in our perception owing to the recursive action of the transformation, and it is a consequence of how one deals with E, seeing it as invariant under T, that makes it into an object for our perception. That E appears as an object is a part and parcel of being an eigenform. Thus, Foerster (2003) spoke of eigenforms in the phrase “objects as tokens for eigenbehaviours”. Heinz, in a wonderful turn of perception, turned the mathematical idea on its head. He pointed out that ordinary objects are tokens for eigenbehaviours. *Ordinary objects are invariances of processes performed in the space of our experience.* The “space of our experience” is the context in which we have our experience and it is the experience itself. I make objects by finding fixed points in the recursion of my interactions. Eigenforms are a touchstone for the relationship of circular and recursive processes and the ground of our apparent worlds of perception. This point of view should be compared with that of Maturana and Varela (1987), in which they see the construction of conversational domains through the coordination of coordinations of actions of organisms in the course of their evolution.

At this point, we have made a special language for our patternplay by borrowing a functional notation from the language of mathematics. We have allowed ourselves to write $T(x) = [x]$ and understood that this is a rule that accepts a given pattern x and allows us to enclose that pattern with brackets. Thus, we could write $T(\#\$\#) = [\#\$\#]$. *And we can ask the following question*:

Q1. Is there any finite sequence of symbols x such that $x = [x]$, where equality means that the symbol sequences are identical?

Well, the answer is no! And the reason why it is no is that $[x]$ has two more symbols in it than x does. It has extra symbols [and]. Therefore, there can be no finite solution to the equation $x = [x]$ if we take equality in this strict way. This is a mathematical argument and it allows us to focus on the important question:

Q2. What is the meaning of equality?

At this point, I hope the reader can begin to see why I want her to pay attention to the logical, mathematical level of the cybernetics. A question that begins in the ground of simple literal equality has an enormous reach. On the one hand, it is not so simple to know just what it means for two strings of symbols to be exactly identical. I will not even try to talk about that. But it is also the case that equality is used in a wider sense where we say, for example, that all men are created equal, and this does not mean that all men are created identical. It means that extra distinctions among men and women, extra discriminations that might block their freedom, are to be avoided. And how we are to avoid them and how we are to achieve tolerance and equality in this general domain is a deep problem to which our cybernetic musings may contribute. The contribution will come from an appreciation that cybernetics has its roots in patternplay and the circular nature of our discourse and design that goes beyond taking the mathematics in a strictly literal mode. In this sense, we must go forward and beyond into new mathematics that can carry the flexibility that we need. I propose that eigenform is part of the beginning of such an expansion of the concept of mathematics and distinction, description, recursion and design.

There is another way to make the fixed point for $T(x) = [x]$. Define a new operator G so that:

$$Gx = [xx]$$

That is you insert two copies of x and surround them by brackets. For now, think of this as a game with letters. But then replace x by G and you see $GG = [GG]$ and we have the GG is a fixed point for T. The reader will note that we seem to have accomplished the impossible! According to the previous paragraph, there can be no finite way to find a fixed point for T. Or argument was that $[GG]$ should have two more symbols than G, and so cannot be equal to G. But we have done it! What does this mean? *It means that we have shifted the notion of equality*, perhaps you did not notice it. First of all, we said G was an operator and so no one told you how many symbols it comprised. Then $Gx = [xx]$ does not mean that the symbols on the left count the same as the symbols on the right. $Gx = [xx]$ is an injunction or a recipe. G applied to x, duplicates x and puts it in the brackets. Thus $G* = [**]$. The expression on the right has four symbols. The expression on the left has two symbols (if you think that G is just a symbol). But $G* = [**]$ makes sense. It means $[**]$ is *produced* by G when G meets *. Then we see that $GG = [GG]$ means that $[GG]$ is produced by GG. Aha! GG is autonomous.

GG produces a new form [GG] from itself. GG has time in its form. GG produces [GG] which produces [[GG]] which produces [[[GG]]] *ad infinitum*. And we have been returned to infinity as an endless state of time. Without this framework of patternplay, we would be shuttered away from the roots of process and thought that are inherent in cybernetics. We must not lose the connection to the mathematics, to the patternplay. We must go beyond conventional language and the *status quo* into new improvisations of pattern and design in a world where our participation is central to the creation of our universe.

At this stage, an eigenform is a game of substitutions, and it is related to the idea of reflexivity. The idea of reflexivity is very simple and it provides most of the examples that make the theory of eigenforms palpable and not just abstract mathematical. However, the idea of reflexivity involves regarding a world outside the prevailing paradigm. To prepare for reflexivity, you must include yourself in the world. You must give up, at least for the moment, the idea that there is a world completely independent of you. You must go back to the young child's perspective and see that everything that you sense and know is involved with your own awareness. I say involved, because I do not assert that you create the whole world. I assert that anything that you know directly is known through your awareness. To come to ground with eigenforms, recognize that you are in a living world populated with actors such as yourself and that the primary entities in that world, the elementary particles, if you will, are persons, actors, awarenesses. And these actors are each both objects (for themselves and for others) and they are each transformers of the entire world of persons. This is the reflexive world. The beginning of this paragraph is a proof that every actor in such a reflexive world has a fixed point, an eigenform. That eigenform for the actor T is created through the persona G who being aware of any x, produces the awareness of T being aware of the self-awareness of x. That is, G is the awareness that T can have of other beings being self-aware. This awareness is the glue that produces the eigenform GG of T being aware of GG herself. It is a key tangle of self and other reference that forms the glue for the reflexive domain of selves that we truly bring forth in our mutuality. It is this mutuality and the context of mutuality to which we dedicate this paper. Moving and transforming in the reflexive world, the experience of eigenforms arising and dissolving is the state of living being.

Reflexive domains

A *reflexive domain* is an abstract description of a conversational domain in which cybernetics can occur. Each participant in the reflexive domain is also an actor who transforms that domain. In full reflexivity, each participant is entirely determined by how he or she acts in the domain, and the domain is entirely determined by its participants. I write the following equation to denote this reflexivity of a domain D (Kauffman, 1987, 2001, 2004, 2005, 2009, 2012a, 2012b, 2015; Scott, 1971; Varela, 1979)[1]:

$$D = [D, D]$$

The symbol $[D, D]$ denotes all the available transformations of the domain D to itself. The equation says that D is identical to the processes that transform it. The point to note is that if a transformation T is defined by the equation $T(D) = [D, D]$, then the transformation T applied to a domain D reveals all the processes that can transform D into itself. *A reflexive domain is itself an eigenform:* $D = T(D) = [D, D]$. Note that a reflexive domain is a context for action, and when we say that the domain D is itself an eigenform, we stand back momentarily from the domain into a larger context that can include it. This means that no domain, even a reflexive domain, is the end of our deliberations. Each domain can be transcended to a new and larger domain. The process is endless and is the source of all our constructions and considerations.

With this in mind, look again at the defining property of the reflexive domain D, $D = [D, D]$, and understand that *the domain itself is an eigenform*.

This structural observation has consequences for the applications of cybernetics and applications to the epistemology of second-order science (Kauffman, 2015; Müller and Riegler 2014; Umpleby, 2014). For if science is to be performed in a reflexive domain, then one must recognize the actions of the persons in the domain. Persons and their actions are not separate. If an action is a scientific theory about the domain, then this theory becomes a (new) transformation of the domain. Theory inevitably affects the ground that it studies. Furthermore, the fact that an entire domain can be seen as an eigenform suggests that one can be an observer of that domain in a wider view of the landscape. Thus physics can be seen as a reflexive domain and one can take a meta-scientific view, allowing physics itself to be one of the objects of a larger domain of in which it (physical science) is one of the eigenforms.

How then, do physical and natural science manage to obtain their apparently objective results? The answer lives in circularity and eigenform. Traditional science searches only for those actions that are independent of the observers (persons) involved. This means that traditional science asks to work within a particular subset of available eigenforms. Once this is realized one can begin to see how to widen scientific operations.

The point I want to make here is that the initial place to begin cybernetics and second-order science is in the recognition of circularity and reflexivity.

The subsequent sections discuss and lead outward to the structure of science and the meaning of eigenform and reflexive domains. I show that our concept of reflexive domain is itself an eigenform and that the central, circular concept of eigenform underlies and overlies all notions of reflexivity. In the Conclusion, I discuss science and second-order science directly. It is my thesis that all science, all attempts to find knowledge, are faithfully modeled by the search for eigenforms in a reflexive domain. A conventionally successful science will find such forms and point to them as the objective results of that science. But a wider look at the situation will reveal that the larger landscape of the reflexive domain has been significantly influenced by these theories. The world has changed as a result of the scientific activity. There is no inviolate ground. The path is being constructed in the act of making it. And yet there are beautiful and important eigenforms to be found. The quest of science becomes larger and more romantic and more intentional in the embracing of the second-order cybernetics. I imagine worlds and bring them into being.

Cybernetics and reflexive domains

I should note that once cybernetics is defined in terms of itself, it becomes what is commonly called "second-order cybernetics". Perhaps there never was first-order cybernetics, but this is a useful distinction. When I distance myself from an object of study, but nevertheless, examine its circularities, it may be natural to call such a study cybernetics of the first order. Here I am concerned with the second order, but I will just call that cybernetics.

There is a way to do mathematics that is explicitly circular, called Lambda Calculus (Barendregt, 1985; Kauffman and Buliga, 2014; Kauffman, 2015; Scott, 1971). It is called "Laws of Form" (Spencer-Brown, 1969) in its full simplicity.

First I want to discuss Laws of Form and then pass over to more complex versions.

In George Spencer-Brown's book *Laws of Form* (Spencer-Brown, 1969), a very simple mathematical system is constructed on the basis of a single sign, the mark, designated by a circle or a box or a right angle bracket. I shall not develop the formalism here. But that sign, in our eyes, makes a distinction in plane in which it is drawn. And the interpretation of Spencer-Brown's calculus has us understand that the sign refers to the distinction that the sign makes (we make that distinction when we are identified with the sign). Thus the sign of distinction in the calculus of Spencer-Brown is self-referential. That is, expressions in the calculus refer to themselves and to other expressions in the calculus.

While this calculus is circular and self-referential, it is consistent.

By a series of departures and constructions, the calculus of indications of Spencer-Brown can be seen to be a foundation for the construction of all of mathematics, and it gives direct insight into the structure of language and communication. The external observer is intimately involved in this calculus, for it is through that observer that the distinctions take place. Thus, the circularity of the calculus engulfs the observer. One is in a cybernetic domain.

A more complex form of circularity in mathematics occurs when an expression *explicitly* refers to itself. For example, I could write the self-referring equation $x = 1 + 1/x$ and have an expression x that is defined in terms of itself. More general fixed points indicate other circularities. The following example is a sentence that refers to itself and tells a truth about itself.:

"This sentence has thirty three letters."

I can formalize circularity in terms of fixed points. Foerster (2003) was fond of this relationship of circularity and fixed points. He saw that a fixed point, taken in the cognitive world, can connote an object in the perception of an observer. In his terms, the object is a token for an eigenbehavior, the iteration of a transformation whose fixed point is the object. The observer embodies a transformation T that leaves the object fixed in the sense that one has the fixed-point equation T(Object) = Object. This means that no extra distinction is introduced by the application of the transformation T to the Object. Von Foerster called the object so fixed as an *eigenform*, and the transformation as an *eigenbehavior*.

It should be understood that the object, the eigenform, may come into existence through the action of the transformation. *This can happen by an interlock of meanings without any iteration.* By an interlock I mean a direct reference of the sentence to itself or a mutual reference of two structures. For example, the US dollar bill has a statement written upon it that says "This bill is legal tender". That statement interlocks with the dollar bill upon which it is printed. When we shake hands, each hand grasps the other hand. The hands interlock. In the next example, we have the sentence "I am the one who says I". The sentence prior to the one you are now reading shakes hands with its reader and makes an opportunity for the reader to identify himself or herself.

I have the following operator:

$$T(x) = \text{the one who says } x$$

The eigenform for this operator is "I".

"I am the one who says I"

"I = the one who says I"

$$\mathrm{I} = T(\mathrm{I})$$

This explication of "I" as an eigenform places its fundamental circularity starkly before us. I is an eigenform of the transformation, but the identity of the I can vary. I can be the one who says I and so can you. The eigenform I is fixed by a linguistic transformation but is handed back and forth in the conversation among a community of I's. This is exactly the condition that Spencer-Brown (1969, p. 76) describes for his mark when he says "We see now that the first distinction, the mark and the observer are not only interchangeable, but in the form, identical".

A community of observer/participators forms a *reflexive domain* D. By this term, I mean that each person in the domain is also an actor in that domain. Each one acts upon the others and each can be acted upon by the others and by himself. This is the fundamental condition for the cybernetics of social and economic behavior, and it is also the condition for all scientific behavior.

There is a key relationship between reflexive domains and eigenforms. I state this relationship as a theorem.

Church–Curry Fixed Point Theorem

Let D be a reflexive domain and let T be any element of the domain D. Then there is an element X in D such that X is a fixed point for T. That is, $TX = X$.

Proof. Define the following operation on elements of D:

$$Gx = T(xx)$$

That is, G operates on x by letting T operate on the result of x operating on itself.

Because D is a reflexive domain, I can assume that the operation G is itself an element of the domain. Now examine the result of G operating on itself. G operating on itself is the result of T operating on G operating on itself. That is, $GG = T(GG)$. Thus, GG is a fixed point for T. *QED*.

It is quite possible that the reader will find this construction, simple, as it is quite baffling! It is best to discuss it from a number of angles. The key point for a reflexive domain is that participants in the domain can act on themselves. One must take this property to heart. Regard yourself as such a participant and understand that you interact not only with the others (for example in a market) but also with yourself. There is a singularity about interacting with oneself that is different from interacting with others. You have to separate yourself from yourself to interact with yourself, and this separation is at a conceptual level or an imaginary level, as you are not actually separate from yourself. Think of the G in the Theorem as a person. This person acts on another person P by letting the transformation T act on the self-interaction of P. What does it mean to act on the self-interaction of another? It means to go into conversation with that other. Thus, G induces T to go into conversation with P. One can think of G as a coach who gets people to interact with one another. But how does G act on himself?

G acting on G induces T to be in conversation with G. This means that G's self-interaction is acted on by T. Thus, G's self-interaction is indeed the action of T on G's self-interaction. The singularity of self-interaction leads to the fixed point.

This singularity of self-interaction is well-known in many puzzles and paradoxes. For example, there is the tale of the village with a Barber B who shaves only those who do not shave themselves. This is straightforward just so long as I do not ask "Who shaves the barber?". For if the barber shaves himself, then by definition, he cannot shave himself.

A deeper relationship with the Fixed Point Theorem is the Russell Paradox. For this purpose, let AB denote the relationship "B is a member of A". Thus A acting on B is the act of B becoming a member of A. Then I can define $Rx = \sim(xx)$ where $\sim$ denotes "not". R is a set with the property that *x is a member of R exactly when x is not a member of x*. R is the set of all sets that are not members of themselves. This set R is the famous Russell set. The paradox of the Russell set is that RR is an eigenform for negation. For I have:

$$Rx = \sim (xx) \text{ and so } RR = \sim (RR)$$

This Russellian eigenform is a problem for those who insist that negation must not have fixed points. Certainly *RR* is a rogue set from the usual point of view since it apparently can both belong to itself and not belong to itself. This depends upon what is the "usual point of view". Looking from the reflexive domain of persons and selves, I see that *RR* is very like a person. *RR* is part of itself, but *RR* is also the whole of herself and so not a part at all. *RR* is a member of *RR* and *RR* is not a member of *RR*. In the reflexive domain of persons and selves it is natural enough for *RR* to be a eigenform of negation.

It is an axiom of physical science that one requires results and procedures that can be repeated by others to a sufficient degree of accuracy so that the community of physicists agrees that the results and procedures do not depend on the specific observers (other than that they carry out the actions of the experiment according to the specifications). It should be clear from this well-known description of experimental physical science that an experiment is a transformation E that involves persons P and a result X and I desire an eigenform X so that $E(P, X) = X$ independent of the choice of P (within a community of "competent scientists" P). Why does X appear on the left-hand side of this equation? By this I mean that the experiment is repeatable. You can record X as a result of the experiment and then find that when you perform it again, you get the same result. It is the repeatability that makes a successful experiment into an eigenform.

That knowledge should be independence of the observers is related to repeatability and I will discuss it further below.

Physics is defined circularly and it demands certain eigenforms for its successful experiments. These eigenforms are in accordance with the description of a world that is objective, a world that is independent of the particular choice of observers. The objective world of physics is not given a priori. The objective world of physics is a construction of the physicists based on the type of results that they deem acceptable as valid physics. Of course it could have happened that physics would fail and there would never be any experiments E of this kind or any results X of this kind. Fortunately for physics there is a whole world of physical results, and it would appear that there is no end in sight to the progress of physics.

I see that one could look at a science that studies eigenforms and reflexive domains and that the participants in such cybernetics are themselves elements/actors in the reflexive domain. One is then led to the question:

Q3. Which is more fundamental, the circularity of eigenform or the reflexivity of the domains in which these eigenforms occur?

It is easier to start with an eigenform, as I have a sense about how to make an eigenform by iterating a transformation. When one does this iteration one learns curious lessons.

One lesson is the astonishing fact that one can make an eigenform just from the form of a transformation as in $X = TTTT\ldots$. Here, X is an eigenform for T, as formally $TX = X$. But is this not some sort of magic trick, a sleight of hand? Indeed it is.

Our usual method produces an infinite limit:

$$X = TTTT\ldots \text{ with } TX = X$$

The fixed point does not depend upon the initial conditions. If I take an example like:

$$Tx = 1 + 1/x$$

then $X = TTT\ldots = 1 + 1/(1 + 1/(1 + \ldots))$ and $X = 1 + 1/X$. I can ask for numbers that satisfy this equation, and there are numerical solutions [the Golden Ratio, Phi = (1 + Sqrt

(5))/2 is a solution to this equation]. This independence of the eigenform on the initial conditions is analogous to an experiment whose results are independent of the particularities of the experimenter.

I would like to make a more precise version of this analogy with the possible independence of observation so that one could consider experiments that are both sensitive and insensitive to the conditions of the experimenter. I describe a scientist to be in search of eigenforms that are independent of the action of the observer. Iteration of a transformation can, as explained in the previous paragraph, erase some of the initial conditions that are characteristic of a particular observer. This is only part of the story. To make a complete account of how a given science strives to make its results observer-independent is a project for future workers in this field of second-order science.

Eigenforms occur within a reflexive domain, giving rise to recursion, and recursion itself brings forth that reflexive domain. The entire reflexive domain is an eigenform. This shows how eigenforms can be dynamic. The entire world of science is itself an eigenform. The world of the stock market is an eigenform. The cybernetics world is an eigenform. These eigenforms are the forms of closure that occur in large interacting systems where the dynamics of individuals leads their identities outward and the individuals become identified with their actions. Just so do words become identified with their actions in Wittgenstein (1958, p. 43): "The meaning of a word is its use [. . .]". In the same way Charles Sanders Pierce (Kauffman, 2001) had the concept of the individual as a sign for himself. This means that in the dynamics of sign-behaviour there arises reference and eventually self-reference and through this self-reference the individual becomes a sign for himself. Word and individuals are both seen as elements of a reflexive domain in the work of Wittgenstein and Pierce. Pierce's concept is directly related to the reflexive domain in which the individual comes to have an identity through the eigenform "I".

With this I return to the identification of the mark and the observer in Spencer-Brown (1969), now at the scale of the detailed interactions of individuals with many others.

I have the Fundamental Theorem that every element T of a reflexive domain D has a fixed point X with $T(X) = X$. Eigenforms prevail at the individual and the global levels in reflexive domains. The world of reflexive domains and their eigenforms is the world in which cybernetics occurs.

How does a reflexive domain come about? In some cases such as our personhood one simply lives there and cannot easily give an account of how it happened. I live in the eigenform of my reflexive domain of self-other interaction. It is a mystery how it happens. I do not have a sequential construction for it. On the other hand, consider some given domain D, not reflexive. By not reflexive I mean that the transformations of D to itself, are not in perfect correspondence to elements of D. There are transformations A taking D to D that are not members of D. What should I do? I may include them and form a new domain D^1. But D^1, while larger than D, may still not be reflexive, so I may need to adjoin more elements. The general pattern is $T(D^n) = [D^n, D^n] = D^{(n+1)}$. Then I iterate the construction of enlarging the domain to infinity or to a very large N where D^N and $D^{(N+1)}$ and indistinguishable. At this point, I have achieved D^N as an eigenform for T. I have created a reflexive domain. Once I have a reflexive domain, I can avail myself of the Church–Curry Fixed Point Theorem within that domain and all the other amenities of closure that such domains afford. The insight that reflexive domains are themselves eigenforms is directly related to how they can come into existence. The technical details of this limit process can be seen in the work of Scott (1971).

With this point of view I can think of a reflexive domain as arising in the freedom to assign names to processes, and to allow those processes to become new elements of the original domain of objects (under scrutiny). This sort of freedom is well-known to computer programmers of languages where an algorithm or a procedure is given a name and that name is then at the same level as other original terms in the language. The language expands in time as the programmer makes new "macros" of this sort. In this practical situation, one does not have to go to the limit of an infinite tower construction (as described in the previous paragraph) to obtain reflexivity. One only has to recognize that defined processes are subject to being elements of an always-extending language.

The construction I made in the Church–Curry Fixed Point Theorem can be accomplished under relatively ordinary circumstances. For example, suppose that I define $Gx = [xx]$ as a *symbolic process*. It is a process that a computer can perform on symbol strings such as x. Then applying G to x means that I duplicate x and place brackets around the two copies of x. This is an operation that can be performed by a computer. I regard G itself as a string of symbols (with one character). The strange and curious thing is what happens when I apply G to G.

With $Gx = [xx]$, I obtain $GG = [GG]$. I obtain a statement of self-reference and, if I take this equality seriously as an action, I obtain the unending recursion (GG is replaced by $[GG]$).

All this is compatible with the previous notion that:

$$E = [[[[[[[\ldots]]]]]]]$$

is invariant under bracketing: $E = [E]$.

In one case, I obtain the eigenform by singular substitution. In the other case, I obtain the eigenform by a limiting process. It should be noted that these examples of strict substitution and transition to infinity are idealized and do not begin the wonderful partial aspects of self-reference that occur in our worlds of language. For example, I may refer to myself without yet knowing what boundaries of the self are intended and indeed with the hope of extending them. I may say "I am a mathematician". or "I am a clarinetist", in each case using the declaration as a performative act that may promote the one who says it into that category. The act of saying can be an essential ingredient in the movement to a state of being. Thus does one say "With this ring I thee wed".

It is at this point that the reader will discern that I am in favor of both nouns and verbs. In fact, from our point of view it is indispensible to have both nouns and verbs and the possibility of changing nouns to verbs and verbs to nouns as the need arises. Owing to the long discussions in systems theory about the nature of nouns and verbs, this aspect of our epistemology will be continued in other papers.

I shall now see how self-reference can arise in systems that are powerful enough to have reference, and powerful enough to take the names of processes and the names of things and use them in the language on an equal basis (just as one does in human natural languages). Nouns and verbs become interchangeable. Suppose that I have a system S that can make distinctions in its world and give names to the sides of these distinctions.

S has a language, and in that language S will write $A \rightarrow B$ to mean "A refers to B" or that "A is the name of B". S is endowed with a special process denoted by "#" so that when $A \rightarrow B$, S creates a new reference $\#A \rightarrow BA$ called the *indicative shift* of the original reference $A \rightarrow B$. Thus $\#A$ is the name of "B with a name-tag A attached".

I say that $\#A$ is the meta-name of BA. This shifting of names to meta-names is fundamental to the observing system S. The shift enables an object and its name to be superimposed just as I find that persons I know have their name as one of their direct properties for me. It is the shift of naming that allows the system S to achieve self-reference. The system S will create a name for the operation of the shift itself.

Let $M \rightarrow \#$ denote the name of the shift operation. Note that this is an instance of a process (#) acquiring a name (M) and thereby being associated with a noun. But then this naming of the shift process can itself be shifted, and I find that $\#M \rightarrow \#M$ is the result of that shift. In this way the system S acquires self-reference. If I let $I = \#M$ then I find that I refers to I. If the system could talk, she would say, "I am the meta-name of my own meta-naming process!". The structure of this acquisition of self-reference has the same pattern as the Church–Curry Fixed Point Theorem. Our point in describing it is to show that the self-reference of "I am the one who says I" occurs naturally in a model of a system *S* that is capable of reference, naming and distinction. Such systems deserve the name of observing systems.

I am now in a position to discuss second-order science in light of an understanding of reflexive domains. I have discussed how a society of persons can be schematically modeled by a reflexive domain *D*. The domain contains more than just the human persons. It contains processes that are initiated by them, with each process regarded both as an object in the domain and as a transformation of the domain. The transformative properties of an element can be referred to any other element or person in the domain. I can now look at the practice of science in terms of the landscape of the domain *D*. I emphasize that the domain *D* should be seen as "the world" in the sense of Wittgenstein. Here I refer to a world that begins in the context of the early Wittgenstein of the *Tractatus* (Wittgenstein, 1922), but it continues into the world of the later Wittgenstein of the *Philosophical Investigations* (Wittgenstein, 1958). From the point of view of reflexive domains, these worlds are not incompatible. What is the case is the present state of action and eigenform. But there is no longer an impenetrable barrier between the descriptions (pictures) of the world and the world itself. Each acts upon the other. Each creates the other. Words acquire their meanings via actions and distinctions occur in the sharp invariances of certain eigenforms. The world is everything that is the case, and the world evolves according to the theories and actions of the participants in that world. Thus when a scientist in the world proposes a theory, this theory becomes a new element of that world. The theory acts and is acted upon by the world of persons and elements of the world. There is no protection for the world from the effects of such a theory. It is remarkable that physicists have been able to isolate phenomena that do not appear to be affected by the theories of those phenomena.

The key to the apparent objectivity of a science is in its carefully crafted eigenforms that are checked to remain fixed throughout significant periods of time. If one attempts to maintain this sort of objectivity in other forms of social science, the processes can take a much different shape. For example, one can design trading algorithms for the stock market on the basis of seemingly reliable information about the behavior of the market. But these very algorithms, when entered into daily practice in the market itself, affect the action of the market and can even lead to instabilities and behaviors that were nowhere in the original theories. Certainly the same remarks apply to the theories and even the opinions of investors, which when made public can affect the action of the market itself. One may say that this indicates that the market is not a subject for objective science, but this is not necessarily the case. Our definition of objective science is that the actions in the reflexive domain produce relatively stable eigenforms. Thus a new version of stock market economics could arise that searches for such regularities even in the face of the publication of the theories themselves.

Other aspects of knowledge certainly have this reflexive pattern. There is a science to learning to play a musical instrument, and it involves principles that do not appear to be changed by the acts of practice. But other aspects such as learning to improvise are clearly a matter of finding a moving eigenform in the course of the action of the play. A simpler example is learning to ride a bicycle. The riding of a bicycle is an eigenform in action and there is no way to find this eigenform except to perform the action. Theories of games that do not submit to exact analysis such as chess or Go evolve in relation to the play of the game. The known theories of chess have seriously affected tournament play, and in turn this tournament play has changed the evolution of the theories.

My thesis is that all attempts to find stable knowledge of the world are attempts to find theories accompanied by eigenforms in the actual reflexivity of the world into which one is thrown. The world itself is affected by the actions of its participants at all levels. One finds out about the nature of the world by acting upon it. The distinctions one makes change and create the world. The world makes those possibilities for distinctions available in terms of our actions. Given this point of view, one can ask, as one should of a theory, whether there is empirical evidence for this idea that stable knowledge is equivalent to the production of eigenforms. In this case, we have only to look at what we do and see that whenever "something is the case" then there is an orchestration of actions that leaves the something invariant, making that something into an eigenform for those actions. The eigenform thesis is not itself a matter of empirical science. It is a matter of definition, albeit circular definition. Another point of view is that the empirical evidence is all around you. Examine any thing. How does it come to be for you? Investigate the question and you will find that thing is maintained by actions. The action could be as simple as opening your eyes and looking at the cloudy sky. With that action, the cloudy sky comes to be for you. I do not assert that this is the usual scientific explanation of cloudy sky. But if you want to work with such things then it is usually even more transparent. The sharp spectral lines of Helium are the result of setting up a very particular experiment that produces them. The experiment, its equipment, the scientists and all that is needed to perform it is the transformation whose eigenform is the spectrum of Helium.

In the last paragraph I suggested that an eigenform was not an empirical matter, but rather a matter of definition. But now I have to assert that definitions are empirical matters! How do we come to agree upon a definition or a procedure? We come to agree through dialogue and through the following of each others injunctions. There is no guarantee that the agreement on the definition will come forth. There is no guarantee at all. We are fortunate when that happens and it is that good fortune of finding common ground that we celebrate in eigenform. In that regard everything that is said is an invitation to possible agreement based on empirical exploration. This exploration includes us, the explorers, as we begin to find the territory that includes us in our creation.

It is a fruitful beginning to look at present scientific endeavors and to see how they are interrelated and find connections among them, to engage in meta-scientific activity. This can reveal how theories, seemingly objective, actually affect the world through their very being, and how these actions on the world come to affect the theories themselves. In exploring the world, we find regularities. It is possible that these regularities are our own footprint. In the end we shall begin to understand the mystery of the eigenforms that we have created, constructed and found.

Note

1. These references deal with both Lambda Calculus and the notion of reflexive domains. In this listing I have indicated only those references that explicitly or implicitly deal with reflexive domains. A reflexive domain is a particular situation where Lambda Calculus applies.

References

Barendregt, H.P. (1985), *The Lambda Calculus, Its Syntax and Semantic*, North Holland, Amsterdam.

Foerster, V.H. (2003), *Understanding Understanding: Essays on Cybernetics and Cognition*, Springer, New York, NY.

Kauffman, L.H. (1987), "Self-reference and recursive forms", *Journal of Social and Biological Structures*, Vol. 10 No. 1, pp. 53-72, available at: http://cepa.info/1816

Kauffman, L.H. (2001), "The mathematics of Charles Sanders Peirce", *Cybernetics and Human Knowing*, Vol. 8 Nos 1/2, pp. 79-110, available at: http://cepa.info/1823

Kauffman, L.H. (2004), "Virtual logic: fragments of the void – selecta", *Cybernetics and Human Knowing*, Vol. 11 No. 1, pp. 99-107.

Kauffman, L.H. (2005), "EigenForm", *Kybernetes*, Vol. 34 Nos 1/2, pp. 129-150, available at: http://cepa.info/1271

Kauffman, L.H. (2009), "Reflexivity and eigenform: the shape of process", *Constructivist Foundation*, Vol. 4 No. 3, pp. 121-137, available at: www.constructivist.info/4/3/121

Kauffman, L.H. (2012a), "Eigenforms, discrete processes and quantum processes", *Journal of Physics: Conference Series*, Vol. 361 No. 1, p. 012034, available at: http://iopscience.iop.org/article/10.1088/1742-6596/361/1/012034/pdf

Kauffman, L.H. (2012b), "The Russell operator", *Constructivist Foundations*, Vol. 7 No. 2, pp. 112-115, available at: www.constructivist.info/7/2/112

Kauffman, L.H. (2015), "Self-reference, biologic and the structure of reproduction", *Progress in Biophysics and Molecular Biology*, Vol. 119 No. 3, pp. 382-409, available at: www.sciencedirect.com/science/article/pii/S0079610715000905

Kauffman, L.H. and Buliga, M. (2014), "Chemlambda, universality and self-multiplication", in Sayama, H., Rieffel, J., Risi, S., Doursat, R. and Lipson, H. (Eds), *Artificial life 14: Proceedings of The Fourteenth International Conference on the Synthesis and Simulation of Living Systems, MIT Press, Cambridge, MA*, pp. 490-497.

Maturana, H.R. and Varela, F.J. (1987), *The Tree of Knowledge: The Biological Roots of Human Understanding*, Shambhala, Boston.

Müller, K.H. and Riegler, A. (2014), "Second-order science: a vast and largely unexplored science frontier", *Constructivist Foundations*, Vol. 10 No. 1, pp. 7-15, available at: www.constructivist.info/10/1/007

Scott, D. (1971), "Continuous lattices", Programming Research Group, Oxford University, Technical Monograph PRG-7.

Spencer-Brown, G. (1969), *Laws of Form*, Julian Press, New York, NY.

Umpleby, S.A. (2014), "Second-order science: logic, strategies, methods", *Constructivist Foundations*, Vol. 10 No. 1, pp. 16-23, available at: http://constructivist.info/10/1/016

Varela, F.J. (1979), *Principles of Biological Autonomy*, North Holland, New York, NY.

Wittgenstein, L. (1922), *Tractatus Logico-Philosophicus*, Harcourt, Brace and Company, New York, NY, available at: www.gutenberg.org/ebooks/5740

Wittgenstein, L. (1958), *Philosophical Investigations*, MacMillan, New York, NY, Originally published in 1953.

Further reading

Alpert, M.I. (2013), "A part of 'a part of': subobject classifiers and mereologies", AB Thesis, Harvard College, Cambridge, MA.

Davies, P.C.W. and Brown, J.R. (1986), *The Ghost in The Atom*, Cambridge University Press, Cambridge.

Russell, B.R. (1938), *The Principles of Mathematics*, W. W. Norton, New York, NY, Originally published in 1903.

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